DAURES GREEN HYDROGEN CONSORTIUM

DAURES GREEN HYDROGEN VILLAGE

University of Stuttgart
IER Institute of Energy Economics and Rational Energy Use

With funding from the:
Federal Ministry of Research, Technology and Space

# Macroeconomic model: NEWAGE-Namibia

## Model characteristics

NEWAGE-Namibia is a recursive-dynamic CGE model.

Selected data sources are:

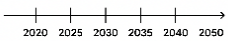
- Global Trade Analysis Project 11
- World Energy Outlook 2024
- International Energy Agency
- International Labour Organization

In a CGE model production factors are used for sectoral production. Government and households constitute the representative agent who can either spend its income on consumption or investment. Production and consumption side are linked through factor and goods markets.

For a more detailed explanation see Fact Sheet "CGE Modelling: Theoretical background".

**Model dynamics:**

The model base year is 2017. The modelling horizon is from 2020 until 2050 in 5-year intervals.

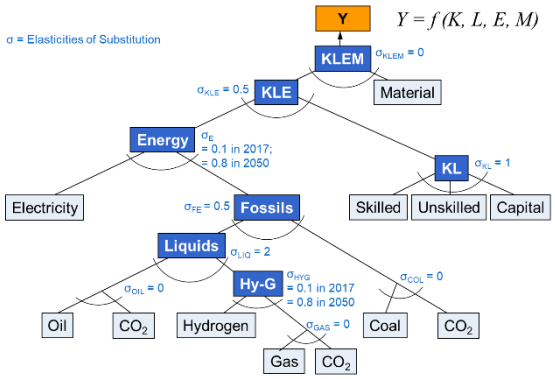



Exogenous increases in labour force and labour productivity over time are growth drivers within the model.

## Production and consumption

Energy related CO2 emissions are linked to energy production from fossil resources.

Production and consumption are modeled through Constant Elasticity of Substitution functions. An exemplary nesting of such a production function can be depicted as follows:




Production factors are skilled and unskilled labour, capital and resources.

## Contact

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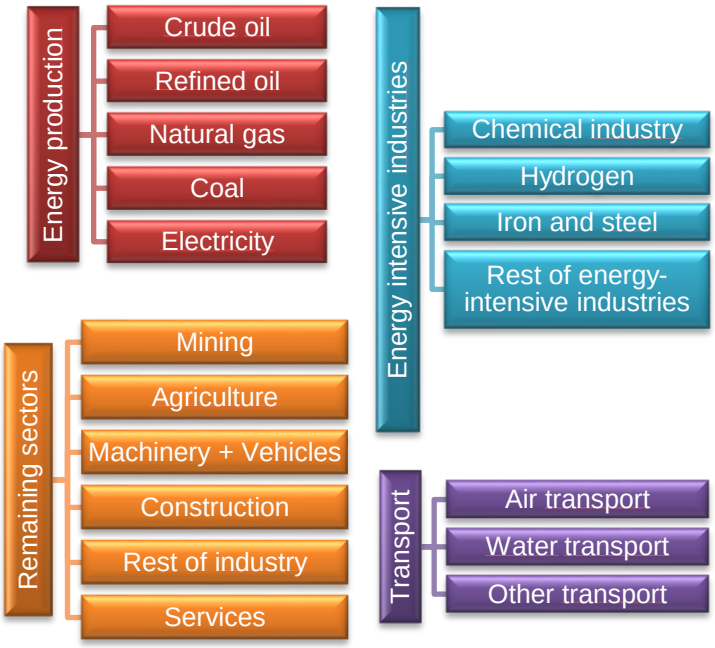
Have a look at our other Fact Sheets covering topics from Project Descriptions, Use Cases, Tech-no-Economic analysis, Indicators, Energy System Analysis, Macro-Economic modelling, and many more:

https://github.com/IER-Hy4Daures/Fact-Sheets

## Sectoral resolution

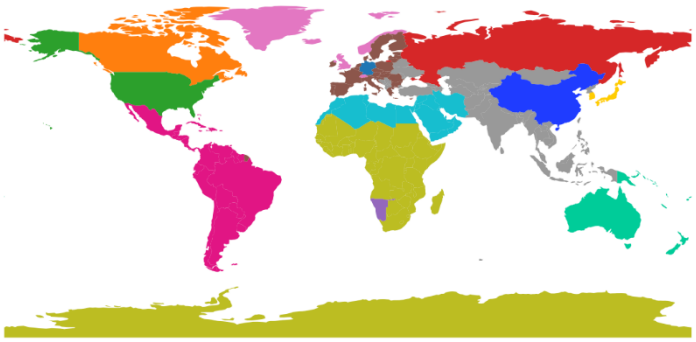
The production factors are utilized for generating industry output within 25 different production sectors, which can be categorized as follows:




Hydrogen can be produced via 5 different production routes:

Electrolysis, steam methane reforming w/o CCS and coal gasification w/o CCS.

Hydrogen can be used together with clean carbon (produced in the agriculture sector) to produce e-fuels and synthetic gas.

## Regional resolution

The model comprises 14 world regions.